Rebuilding the ECB Euro Area Yield Curve

Gianni Pilotti

1 The Core Idea

The ECB publishes two things each day: the six parameters of a Svensson curve, and the spot, forward and par yield curves computed from them. If the formulas are right, rebuilding the rates from the parameters gives the published numbers exactly. That is a clean test of a curve implementation, and a curve that passes it can be used with some confidence for risk and relative value.

- **Bond risk:** price a bullet bond off the curve and measure its DV01 and its key rate durations, so that a hedge is placed where the risk is.
- **Factor structure and relative value:** a principal component analysis (PCA) of daily curve moves, then butterflies that carry no level or slope exposure, tested in a backtest after costs.

The data are the ECB curves from September 2004 to August 2026.

2 Why It Matters

- **Conventions fail silently.** Treating the ECB rates as annually compounded instead of continuously compounded moves the 10-year rate by 6.9 bp on the latest curve. Dividing by a coupon schedule instead of a continuous annuity moves the par yield by 6.8 bp. Both are larger than the 0.5 bp round-trip cost the backtest charges, and neither raises an error.
- **A parallel DV01 hides where the risk sits.** A 30-year bond with a 2% coupon has an effective duration of 20.11 years, and 16.17 of them sit in the 30-year bucket. A 10-year hedge covers 3.14 years. The book is flat to a parallel shift and still loses on a steepening.
- **A few factors drive the curve.** Three principal components explain 96.0% of the variance of daily changes in sample and 98.6% out of sample. Hedging level, slope and curvature removes most of the risk, and what is left is small.
- **A good-looking spread is not a trade.** A butterfly on a fitted curve has no bid-offer and no financing. After a flat cost it earns a portfolio Sharpe of 0.16, with no t-statistic above 2.

3 Assumptions

Every item below is built into the code. The last sentence of each says what moves if it is wrong.

- **Data.** The ECB `YC` dataset, all-issuers and AAA curves, 5,614 daily curves from 2004-09-06 to 2026-08-21, 120 maturities (3, 6 and 9 months, then every 3 months to 30 years). The curves are fitted values, not traded prices. A butterfly P&L on fitted rates may not be capturable with real bonds.
- **Rate convention.** Rates are continuously compounded and in percent (series `SV_C_YM`). Another compounding shifts every spot rate by about $y^2/2$, 6.9 bp at 10 years on the latest curve.
- **Par yield.** Defined against a continuous annuity, with a monthly sum of discount factors before 19 May 2023. Using one convention for the whole history misprices the par yield by up to 1.28 bp.
- **Bonds.** Bullet bonds with annual coupons, cash flows at exact year fractions, no calendar, no day count, no settlement lag, discounted on the government zero curve with no extra credit or liquidity spread. A real bond trades rich or cheap to the fitted curve, so its price differs from the model price by that residual.
- **Risk shocks.** A ± 1 bp parallel shift through β_0, central difference. Key rate shocks use triangular weights on 0.5, 2, 5, 10 and 30 years, flat beyond the ends. A shock of another size or shape changes the convexity term, and any risk beyond 30 years lands in the 30-year bucket.
- **PCA.** A covariance PCA of daily spot changes in bp, loadings fitted on the first half of the dates and scored on the second half. It assumes the factor structure is stable. A regime change, such as the end of negative rates, rotates the loadings and lowers the out-of-sample share.
- **Butterfly neutrality.** Weights are neutral to the first two factors only, from a trailing 250-day PCA, re-solved every 21 days and held in between, and a reset is skipped, keeping the old weights, when the new gross weight exceeds 6.0. Between resets the loadings drift, so the structure carries a residual slope or level exposure.
- **Signal and execution.** A z-score against the trailing 120-day mean and standard deviation, scored at today's weights, with entry at $|z| > 1.5$ and exit at $|z| < 0.5$. The position taken on day t earns the move from t to $t+1$ at the fitted curve, so execution is instant and at mid. Costs are a flat 0.5 bp round trip per unit of DV01, with no financing or repo and no widening in stress.
- **Statistics.** Sharpe is annualised with 256 trading days and the t-statistic is the Sharpe times the square root of the years in the sample. This assumes independent daily P&L: positive autocorrelation would overstate the t-statistic. The five structures are tested without a multiple-testing correction.

4 Method

4.1 Svensson Curve

With $\beta_0, \ldots, \beta_3, \tau_1, \tau_2$ the ECB parameters, the code (`eyc/svensson.py`) uses continuously compounded rates in percent. With $x_i = t/\tau_i$:

$$f(t) = \beta_0 + \beta_1 e^{-x_1} + \beta_2 x_1 e^{-x_1} + \beta_3 x_2 e^{-x_2}, \tag{1}$$

$$y(t) = \beta_0 + \beta_1 \frac{1 - e^{-x_1}}{x_1} + \beta_2 \Big(\frac{1 - e^{-x_1}}{x_1} - e^{-x_1} \Big) + \beta_3 \Big(\frac{1 - e^{-x_2}}{x_2} - e^{-x_2} \Big), \tag{2}$$

$$DF(t) = e^{-y(t)\,t}. \tag{3}$$

The spot rate y is the average of the forward rate f over $[0, t]$. The parameters have a reading: β_0 is the long-run level (the limit of y as $t \to \infty$), $\beta_0 + \beta_1$ is the instantaneous short rate, and β_2 and β_3 add one hump each, at maturities set by τ_1 and τ_2. The code bumps β_0 for a parallel shift because β_0 enters every maturity with a coefficient of one.

4.2 Par Yield

The ECB defines the par yield against a continuous annuity:

$$y_{\text{par}}(T) = 100 \cdot \frac{1 - DF(T)}{\int_0^T DF(t)\,dt}. \tag{4}$$

The integral is evaluated with 64-point Gauss–Legendre quadrature. Before 19 May 2023 the ECB used a right Riemann sum of DF at monthly steps, which understates the integral because DF falls, and so overstates the par yield. The ECB did not restate the history, so the code implements both and picks one by date. The gap grows with the level of rates.

4.3 Bond Risk

A bond's dirty price is the sum of its cash flows discounted with DF. Curve risk uses a ± 1 bp parallel shift, applied exactly by bumping β_0, with $P_\downarrow$ and $P_\uparrow$ the prices after the curve falls or rises by 1 bp:

$$\text{DV01} = \frac{P_\downarrow - P_\uparrow}{2}, \qquad D_{\text{eff}} = \frac{P_\downarrow - P_\uparrow}{2P \cdot 10^{-4}}, \qquad C_{\text{eff}} = \frac{P_\uparrow + P_\downarrow - 2P}{P \cdot (10^{-4})^2}. \tag{5}$$

Key rate durations bump the zero curve with triangular weights on the tenors 0.5, 2, 5, 10 and 30 years. The weights sum to one at every maturity, so the key rate durations add up to D_{eff}, and a test checks this identity. Effective duration is curve based. The yield-based modified duration, which discounts at one redemption yield, is also reported. On a sloped curve the two differ.

4.4 PCA

The PCA uses daily spot changes Δy in bp. Loadings are the eigenvectors of the covariance matrix on the first half of the dates, ordered by eigenvalue λ_k and signed so that each factor loads positively on average:

$$\text{in sample} = \frac{\sum_{k \le 3} \lambda_k}{\sum_k \lambda_k}, \qquad \text{out of sample} = 1 - \frac{\|X - XLL^\top\|^2}{\|X\|^2}, \tag{6}$$

where X is the demeaned second half and L holds the three loadings. Because it uses the covariance, not the correlation, the maturities with the largest daily moves weigh the most.

4.5 Neutral Butterflies

For a butterfly on three maturities with loadings L on level and slope, the weights w solve

$$L_{\text{level}}^\top w = 0, \qquad L_{\text{slope}}^\top w = 0, \tag{7}$$

with the belly weight fixed at one. The solution is the null space of a 2×3 matrix, found with an SVD. The structure level is $\sum_i w_i\, y_i$ in bp, and a long position earns when it falls. Parameters: loadings from a trailing 250-day window, weights reset every 21 days, gross weight $\sum |w_i| \le 6$.

The signal is the z-score of the structure against its trailing 120-day mean and standard deviation, scored at today's weights so that a weight reset does not create a false signal. Because $\mu = w^\top \bar{y}$ is linear and $\sigma^2 = w^\top \mathrm{Cov}(y)\, w$ is quadratic in the rolling moments of the legs, no history needs rebuilding. The position moves between -1, 0 and $+1$ with the entry and exit bands above. The daily P&L is $-\text{position} \times \sum_i w_i(t)\,(y_i(t+1) - y_i(t))$ at constant weights, less a cost of turnover $\times$ gross $\times\, 0.5/2$ in bp. A carry filter exists in the code and is left off: it declines trades that were profitable on average.

5 Method Comparison

Method	Strength	Weakness
Parallel DV01 hedge	One number, one instrument	Blind to slope and curvature
Key rate hedge	Shows where the risk sits	Needs several hedges; tent shape is a choice
PCA-neutral butterfly	Zero level and slope exposure by construction	Weights unstable if legs load alike
Naive 2:−1:−1 butterfly	Simple, fixed weights	Not neutral to level or slope

A parallel DV01 is enough for a short position hedged with a close maturity. A long bond, or any book that spans the curve, needs key rates, because the hedge instrument must sit in the same bucket as the risk. The PCA-neutral weights suit a relative value view on curvature. The naive butterfly stays useful as a check: if a structure only looks neutral, the two series diverge.

6 Results

6.1 Curve Reconstruction

Error against the published rates, in basis points (all-issuers: 5,614 dates, 120 maturities, 673,680 observations).

Curve	Rate	Max error	RMSE	Dates above 0.01 bp
All issuers	Spot	1.5×10^{-8}	1.6×10^{-9}	0
All issuers	Forward	2.0×10^{-8}	1.6×10^{-9}	0
All issuers	Par yield	2.8×10^{-7}	2.7×10^{-9}	0
AAA	Spot	5.6	0.042	1
AAA	Forward	12.6	0.159	3
AAA	Par yield	7.4	0.065	2

The all-issuers curve is rebuilt to numerical precision on every date, for every rate. That fixes the formulas and the compounding convention, and the clean split of the par yield (4,782 dates match the monthly formula, 832 match the integral, none match both) shows the methodology change. The AAA errors come from three dates (2007-05-23, 2020-07-15, 2022-06-23) where the ECB's own AAA parameters and published rates disagree. On two of them τ_1 and τ_2 are almost equal, which makes the Svensson fit unstable. The all-issuers curve is exact on the same dates,

so the error is in the ECB data, not in the code. The test cannot say whether the ECB curve is right about the market, only that it is internally consistent.

6.2 Curve Analysis and Bond Risk

The mean 10-year spread between the all-issuers and AAA curves is 55.4 bp, with a peak of 194.8 bp on 23 July 2012. On the first half of the sample, over all 120 maturities, level explains 74.5% of the variance of daily changes, slope 15.3% and curvature 6.2%. On the 11 tenors of the trading study the shares are 59.1%, 19.2% and 9.7%: they depend on which maturities enter the covariance. The out-of-sample share (98.6%) is above the in-sample one (96.0%). That is allowed, because each is measured against the variance of its own half, and it does not mean the loadings fit better later.

Bond risk on the latest curve (21 August 2026, ECB all-issuers parameters), per 100 face, annual coupons. The yield is the redemption yield, and the last four columns are key rate durations in years:

Bond	Price	Yield	D_{eff}	DV01	0.5y	2y	5y	10y / 30y
2y, 2.5%	99.232	2.90%	1.98	0.0196	0.02	1.96	0	0 / 0
5y, 3.0%	99.039	3.21%	4.71	0.0467	0.02	0.16	4.54	0 / 0
10y, 3.0%	94.116	3.72%	8.72	0.0821	0.02	0.17	0.59	7.94 / 0
30y, 2.0%	61.327	4.33%	20.11	0.1233	0.02	0.17	0.60	3.14 / 16.17

Each of the three shorter bonds keeps 91% or more of its duration in its own bucket. The 30-year bond keeps 80%, because its annual coupons put duration along the whole curve, and a 10-year hedge neutralises only the 3.14 years in its bucket. The key rate durations add to the effective duration (largest gap 2×10^{-6} years). A 10-year duration hedge removes 77% of the variance at 5 years, 39% at 30 years and 3% at 3 months, so most of the remaining risk is slope and curvature. The effective duration (20.11) lies between the modified duration (19.61) and the Macaulay duration (20.46). Modified duration weights every cash flow at one redemption yield and divides by $1 + y$, while effective duration weights each cash flow at its own zero rate, so the two differ by 0.49 years on this bond.

6.3 Butterfly Backtest

Five butterflies, 2005–2026, net of costs.

Structure	Net P&L (bp)	Sharpe	t-stat	Max drawdown (bp)
2s5s10s	108	0.13	0.59	−225
5s10s30s	335	0.30	1.40	−108
3s7s15s	279	0.31	1.43	−107
2s10s30s	168	0.14	0.62	−191
1s3s7s	−305	−0.35	−1.61	−319
Portfolio	117	0.16	0.75	−139

No t-statistic exceeds 2, so the result is not statistically significant. Four of five structures make money and the best two have t-statistics near 1.4, but the 1s3s7s loses 305 bp and its t-statistic of -1.61 is as large as the winners'. The equal-weight portfolio keeps all five (the drop of the worst one was chosen after the fact and is not a result). The portfolio drawdown of 139 bp is larger than its 117 bp of net P&L, so the average gain does not pay for the risk. Costs matter

more than the signal: trading the z-score continuously instead of with entry and exit bands raises turnover from about 11 to about 70 times a year, which makes the 2s5s10s negative.

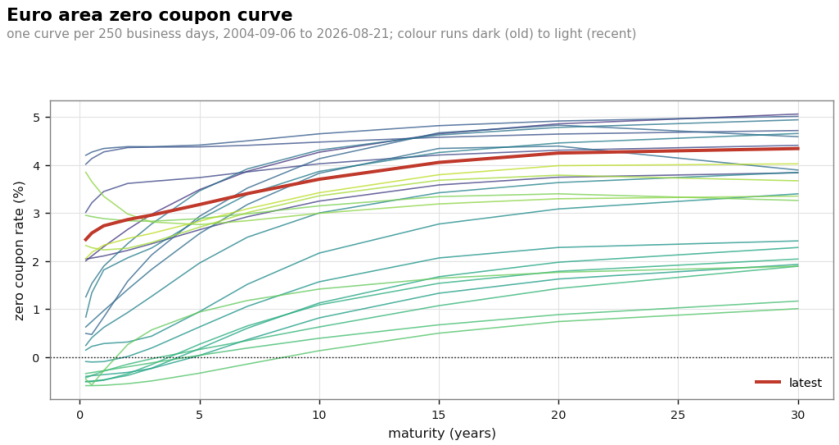


Figure 1: Euro area zero coupon curve, one curve per 250 business days.

7 Worked Example

The Svensson parameters are illustrative: $\beta = (3.0, -1.5, 2.0, -1.0)$, $\tau_1 = 2$, $\tau_2 = 6$.

Spot and discount factor at 10 years. $x_1 = 5$ and $x_2 = 1.6667$. The terms are $3.0 - 0.2980 + 0.3838 - 0.2978$, so $y(10) = 2.7881\%$ and $DF(10) = e^{-0.027881 \cdot 10} = 0.75669$.

Annuity. The Gauss–Legendre integral of $DF(t)$ over $[0, 10]$ is 8.7462.

Par yield. $y_{\text{par}}(10) = 100 \cdot (1 - 0.75669)/8.7462 = 2.7819\%$. The monthly-sum version gives 2.7852%, a gap of 0.32 bp between the two conventions on this curve.

10-year bond. A 2% annual coupon bond has a dirty price of 92.917. Its effective duration is 9.1206, so DV01 $= 92.917 \cdot 9.1206 \cdot 10^{-4} = 0.0847$ per 100 face. The key rate durations are 0.014, 0.114, 0.412 and 8.581 at 0.5, 2, 5 and 10 years, and zero at 30 years. They sum to 9.1206.

Butterfly weights. Take illustrative loadings for the 2, 5 and 10-year points. Level: 0.40, 0.35, 0.30. Slope: -0.60, -0.10, 0.45. With weights a, 1, b the two conditions are:

$$0.40a + 0.35 + 0.30b = 0,$$
$$-0.60a - 0.10 + 0.45b = 0.$$

The first gives $a = -0.875 - 0.75b$. Substituting, $0.425 + 0.9b = 0$, so $b = -0.4722$ and $a = -0.5208$. The code returns the same weights $(-0.5208, 1, -0.4722)$. The structure is flat to level and slope. Its gross DV01 is $0.5208 + 1 + 0.4722 = 1.993$ per unit of belly.

Sensitivity. Moving β_0 from 3.0 to 1.0 and to 5.0 changes the monthly-sum gap to 0.026 bp and 0.952 bp, the 10-year price to 111.62 and 77.50, and the effective duration to 9.21 and 9.02. A higher level means a larger convention gap, a lower price and a slightly shorter duration. For the butterfly, setting the belly slope loading to 0.00 instead of -0.10 moves the wings from $(-0.5208, -0.4722)$ to $(-0.4375, -0.5833)$: a 0.10 change in one loading shifts about 0.1 of weight from one wing to the other, which is why weights are reset on a trailing window and screened by gross DV01.

8 Limits and Extensions

- **Government curve:** it is not a swap or OIS curve, so it cannot discount collateralised derivatives.
- **Parameters not re-estimated:** the euro parameters are the ECB's, because the bond prices behind them are not public. The fitting step is tested on US data only, where the curve is identified but the individual β are not when τ_1 and τ_2 are close (13.5% of days).
- **No calendar:** cash flows are placed at exact year fractions.
- **Curve points, not instruments:** the backtest trades points on a fitted curve with a flat 0.5 bp cost. Real futures or swaps would add basis and financing.
- **No edge shown:** the portfolio t-statistic is 0.75. The backtest is a method check, not evidence of a tradable strategy.

Three extensions would fix the largest limits. First, an OIS or swap curve built from the same code, so that the risk measures apply to collateralised trades. Second, a backtest on the actual instruments (Bund, Schatz and OAT futures or swaps) with financing and bid-offer, which would replace the flat cost. Third, a walk-forward test of the entry and exit bands and the 120-day window, so that the parameters are chosen on one period and judged on the next.

9 Conclusion

Rebuilding a published curve is a test with a right answer, and this one passes to numerical precision. The harder question is what a fitted curve can support: it prices and hedges a bond well, but it does not by itself show a trade, and the gap between a fitted spread and a traded one is the cost the backtest could only guess.